Golden Touchstone: A Comprehensive Bilingual Benchmark for Evaluating Financial Large Language Models

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Abstract

As large language models (LLMs) increasingly permeate the financial sector, there is a pressing need for a standardized method to comprehensively assess their performance. Existing financial benchmarks often suffer from limited language and task coverage, low-quality datasets, and inadequate adaptability for LLM evaluation. To address these limitations, we introduce Golden Touchstone, a comprehensive bilingual benchmark for financial LLMs, encompassing eight core financial NLP tasks in both Chinese and English. Developed from extensive open-source data collection and industry-specific demands, this benchmark thoroughly assesses models' language understanding and generation capabilities. Through comparative analysis of major models such as GPT-4o, Llama3, FinGPT, and FinMA, we reveal their strengths and limitations in processing complex financial information. Additionally, we open-source Touchstone-GPT, a financial LLM trained through continual pre-training and instruction tuning, which demonstrates strong performance on the bilingual benchmark but still has limitations in specific tasks. This research provides a practical evaluation tool for financial LLMs and guides future development and optimization. The source code for Golden Touchstone and model weight of Touchstone-GPT have been made publicly available at https://github.com/IDEA-FinAI/Golden-Touchstone.

1 Introduction

The rapid development of both proprietary (Brown et al., 2020; Ouyang et al., 2022; OpenAI, 2023; Anthropic, 2024; Team et al., 2023) and open-source Large Language Models (LLMs) (Touvron et al., 2023a,b; AI@Meta, 2024; Bai et al., 2023; Yang et al., 2024a; DeepSeek-AI, 2024; Young et al., 2024; Zeng et al., 2023; Baichuan, 2023; Gan et al., 2023; Zhang et al., 2022) has led to their increasing application in various fields, including finance (Wu et al., 2023; Lopez-Lira and Tang, 2023), healthcare (Thirunavukarasu et al., 2023; Tian et al., 2023), and law (Cui et al., 2023; Xiao et al., 2021). Among these, the financial sector shown in Figure.1 stands out as a critical area for LLM application due to its rich textual information and high practical value.

In recent years, a variety of advanced financial large language models (FinLLMs) have emerged, capable of specialized tasks such as financial sentiment analysis, content summarization, stock movement prediction, and question answering (Yang et al., 2023; Xie et al., 2023; Li et al., 2023; Chen et al., 2023; Zhang and Yang, 2023). These models leverage unique frameworks and tuning methods to enhance their performance on domain-specific benchmarks, offering robust solutions for real-world financial applications. However, existing financial benchmarks often suffer from limited language and task coverage, low-quality datasets, and inadequate adaptability for LLM evaluation, leading to poor evaluation results (Shah et al., 2022; Lu et al., 2023; Xie et al., 2023, 2024; Yang et al., 2023; Lei et al., 2023; Zhang et al., 2023).

To address these challenges, we propose Golden Touchstone, a comprehensive bilingual benchmark for financial LLMs, encompassing eight core financial NLP tasks in both Chinese and English. Golden Touchstone provides high-quality datasets, task-aligned metrics, and instructional templates to guide LLMs in generating task-appropriate responses. We evaluated several state-of-the-art models, including GPT-4o, Qwen-3, Llama-3, FinGPT, and FinMA, on this benchmark. Results indicate that while these models perform well on tasks such as sentiment analysis and entity extraction, there is significant room for improvement in areas like

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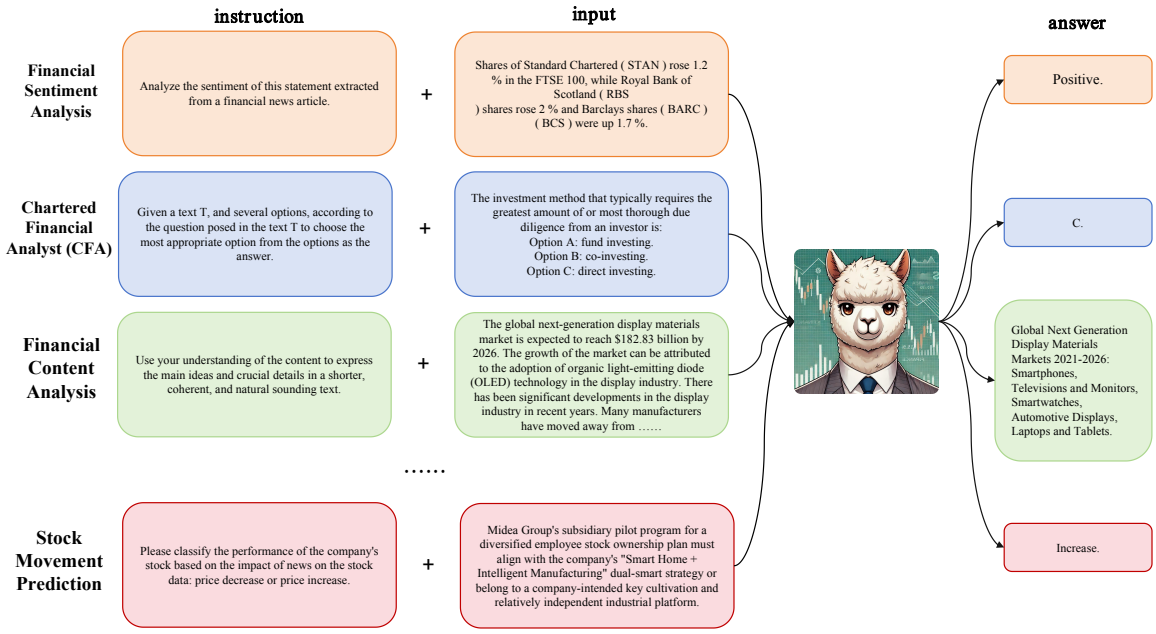

Figure 1: Financial large language models are designed to perform specialized tasks such as financial sentiment analysis, content analysis, stock movement prediction, and financial analyst level question answering by interpreting and processing structured instructions and various input data to generate precise outputs.

stock movement prediction and classification tasks. Additionally, we open-source Touchstone-GPT, a financial LLM trained through domain-specific continual pre-training and instruction tuning, which serves as a new baseline for future research.

Our contributions are as follows:

- Introduction of Golden Touchstone, a comprehensive bilingual benchmark for financial LLMs, encompassing 22 datasets across eight tasks in both Chinese and English.
- Evaluation of state-of-the-art LLMs and Fin-LLMs on Golden Touchstone, highlighting their strengths and limitations across various tasks.
- Open-sourcing of Touchstone-GPT, a financial LLM trained through domain-specific continual pre-training and instruction tuning, fostering further advancements in financial AI.

2 Benchmark Design

2.1 Current Benchmark Status

Existing open-source financial benchmarks have made significant strides in evaluating financial natural language processing (NLP) tasks. FLUE (Shah et al., 2022) pioneered English financial NLP evaluation, covering sentiment analysis, news classification, and other critical tasks. Subsequently, PIXIU (Xie et al., 2023) and FinBen (Xie et al., 2024) expanded task coverage, while in the Chinese domain, BBT-Benchmark (Lu et al., 2023) introduced the first comprehensive Chinese financial evaluation framework.

These benchmarks suffer from several critical limitations. Firstly, there is inconsistent data quality across different tasks, as shown in Table 1. Additionally, large language models face challenges in numerical understanding, a problem highlighted in various studies (Shen et al., 2023; Akhtar et al., 2023; Schwartz et al., 2024; Hu et al., 2024). Lastly, the benchmarks lack bilingual assessment capabilities, a gap that has been pointed out in recent research (Nie et al., 2024; Lei et al., 2023). These limitations collectively hinder the overall effectiveness and reliability of the benchmarks. To address these limitations, we propose a unified benchmark that integrates high-quality financial datasets from both English and Chinese domains. Our approach aims to provide a more comprehensive and linguistically diverse evaluation of financial large language models (LLMs).

2.2 Golden Touchstone Benchmark Framework

Addressing these critical limitations, we introduce the Golden Touchstone benchmark, a comprehen-

Table 1: Diversity of Financial Analysis Tasks Across Different Benchmarks

Benchmarks	Sent. Anal.	Classif.	Ent. Extr.	Rel. Extr.	Multi. Choice	Summ.	Quest. Ans.	Stock Pred.
FinGPT-Bench (Wang et al., 2023)	✓	✓	✓	✓				
FinBen (Xie et al., 2024)	✓	✓	✓	✓		✓	✓	✓
BBT-Fin (Lu et al., 2023)	✓	✓	✓	✓		✓	✓	
Fin-Eval (Zhang et al., 2023)					✓			
CFBenchmark (Lei et al., 2023)	✓	✓	✓			✓	✓	
Golden-Touchstone	✓	✓	✓	✓	✓	✓	✓	✓

sive bilingual evaluation framework designed to holistically assess financial language models. Conceptualized around two primary dimensions—task types and language coverage—our approach represents a significant departure from existing evaluation methodologies. The overview framework can be seen in Figure.2. Firstly, our Golden Touchstone benchmark categorizes financial NLP tasks across two dimensions: the first dimension is task type including Natural Language Understanding (NLU) including sentiment analysis, knowledge examination, information extraction, text understanding tasks and Natural Language Generation (NLG) including summary analysis, question answering tasks, and the second dimension is language including English and Chinese.

The benchmark strategically integrates English and Chinese datasets across eight sophisticated subtasks, spanning Natural Language Understanding (NLU) and Natural Language Generation (NLG). By carefully curating high-quality datasets from existing benchmarks and datasets. The detailed data statistics are shown in Appendix.A. The benchmark encompasses eight critical sub-tasks:

- **Sentiment Analysis**: Utilizing datasets like FPB (Malo et al., 2014) and FiQA-SA (Maia et al., 2018a) and FinFE-CN (Lu et al., 2023)
- **Classification**: Integrating Headlines (Sinha and Khandait, 2021), FOMC (Shah et al., 2023), and LendingClub (Feng et al., 2023) and FinNL-CN (Lu et al., 2023) datasets
- **Entity Recognition**: Using NER (Alvarado et al., 2015) and FinESE-CN (Lu et al., 2023) datasets
- **Relation Extraction**: Incorporating FinRED (Sharma et al., 2022) and FinRE-CN (Lu et al., 2023) datasets
- **Multiple Choice**: Drawing from CFA (Yang et al., 2024b) and FinEval (Zhang et al., 2023) and CPA (Yang et al., 2024b) datasets
- **Summarization**: Employing EDTSUM (Zhou et al., 2021) and FinNA-CN (Lu et al., 2023) datasets
- **Question Answering**: Utilizing FinQA (Chen et al., 2021) and FinQA-CN and FinCQA-CN (Lu et al., 2023) datasets
- **Stock Movement Prediction**: Introducing news-based prediction using CMIN-US and CMIN-CN (Luo et al., 2023) datasets

A implementation consideration involves the treatment of [unknown] labels across various tasks. Our approach consistently treats [unknown] labels as incorrect predictions, which maintains uniform evaluation standards across all tasks, avoids potential biases introduced by alternative methods, and reflects real-world scenarios where models must make definitive predictions. This approach proves superior to alternative methods such as sample omission, random assignment, or fixed selection, which may introduce evaluation biases or distort performance metrics. The benchmark also design incorporates several features to ensure meaningful cross-lingual comparisons, including maintaining identical task structures across English and Chinese benchmarks, using comparable evaluation metrics for equivalent tasks, and choosing datasets with similar characteristics and difficulty levels. These design choices enable robust comparison of model performance across languages while accounting for linguistic and cultural differences in financial contexts.

By addressing previous benchmarks' limitations, our approach provides a more robust, comprehensive, and linguistically diverse framework for evaluating financial large language models. The benchmark not only expands task coverage but also addresses critical challenges in current financial NLP evaluation methodologies. Our key methodological innovations reflect significant efforts in data selection, cleaning, and organization. We replaced time-series tabular data with news-based stock prediction to enhance the relevance and diversity of

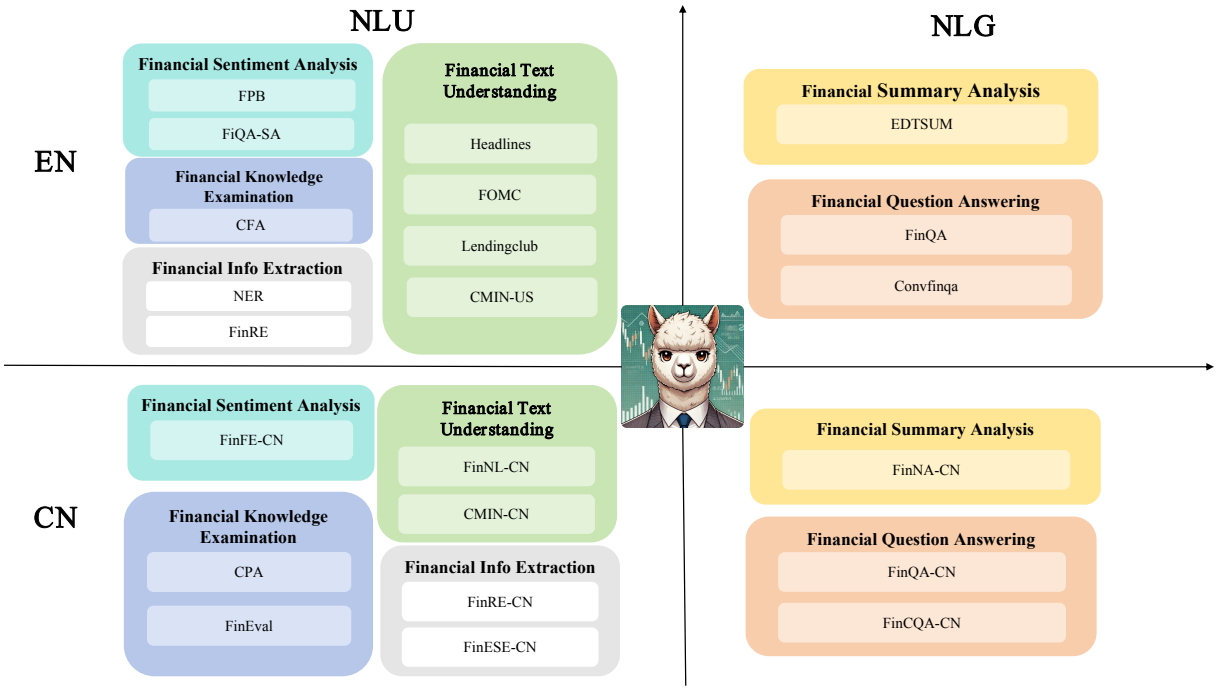

Figure 2: Financial NLP tasks are categorized along two dimensions: task types, divided into financial NLU (Natural Language Understanding) and financial NLG (Natural Language Generation), and language, categorized as English and Chinese. We organized the collected high-quality datasets along these axes.

the dataset. Additionally, we ensured bilingual task alignment to address the lack of multilingual assessment capabilities, while carefully selecting datasets with consistent and high-quality labeling to overcome issues related to data quality. Furthermore, we developed a unified evaluation approach across different financial NLP tasks to ensure robust and comparable results. In line with our commitment to fostering research transparency and collaboration, we have not only open-sourced the benchmark datasets but also released a well-documented, clear, reproducible, and extensible evaluation code framework for the broader research community to utilize and build upon. The Golden Touchstone benchmark represents a significant step forward in financial NLP evaluation, addressing key limitations of existing benchmarks while providing a comprehensive, bilingual framework for assessing model performance. Its systematic design, rigorous evaluation methodology, and practical focus make it an essential tool for researchers and practitioners in financial NLP.

3 Experiments

3.1 Experimental Setup

Baselines. We conducted an extensive experimental evaluation against the Golden Touchstone Benchmark, incorporating a comprehensive array of models. For all models and inference tasks, we set the PyTorch and CUDA random seeds and configured the model with a greedy decoding strategy. This ensures reproducibility of experimental results and eliminates the influence of sampling decoding strategies on the final generated outputs. This included cutting-edge commercial models such as GPT-4o (OpenAI, 2023), alongside prominent open-source alternatives like Meta Llama-3 (AI@Meta, 2024) and Alibaba Qwen-3 (Yang et al., 2025). Additionally, we integrated the latest and most influential financial language models (FInLLMs), namely FinGPT (Yang et al., 2023), FinMA (Xie et al., 2024), CFGPT (Li et al., 2023), and DISC-FinLLM (Chen et al., 2023). These models were meticulously selected to represent a diverse spectrum of capabilities, ranging from general-purpose language understanding to specialized financial domain expertise. Our experiments aimed to rigorously assess the performance, robustness, and adaptability of each model within the context of financial data processing and analysis. The results provide valuable insights into the strengths and limitations of current state-of-the-art models, offering a foundation for future advancements in financial language modeling.

Touchstone-GPT Training. To further con-

tribute to the research and development of FIn-LLMs and Financial benchmarks for LLMs, we have meticulously trained and open-sourced a Touchstone-GPT model. This initiative aims to serve as a valuable resource for advancing the field, providing a robust and versatile model that can be utilized for a wide range of financial language tasks. We adopted a two-stage training strategy comprising continuous pre-training and post-training, based on the Qwen-2.5 (Yang et al., 2024a) foundational model. During the continuous pre-training phase, we initially conducted pre-training on a high-quality financial corpus containing 100 billions tokens, which included textbooks, encyclopedias, research reports, news articles, and real-time analysis, all meticulously cleaned. In the post-training phase, we employed a standard instruction fine-tuning strategy, collecting, cleaning, and formatting a high-quality dataset of 300,000 instruction-response pairs shown in Tabel.4 and Table.5. To avoid catastrophic forgetting in general tasks, we also incorporated general-domain pre-training corpora (Gan et al., 2023) and instruction-tuning corpora (Peng et al., 2023) into continuous pre-training and post-training. This culminated in the final Touchstone-GPT model. We utilized Megatron(Shoeybi et al., 2019) for continuous pre-training and LlamaFactory(Zheng et al., 2024) for instruction post-training as our training frameworks, respectively. In this study, we employ an advanced model training setup using the AdamW optimizer (Kingma and Ba, 2014) with a learning rate of 1.0e-5, cosine annealing scheduler (Szegedy et al., 2016), and a 10% warmup ratio (He et al., 2016; Goyal et al., 2017) to enhance training stability and convergence. We enable gradient accumulation (Shoeybi et al., 2019) and checkpointing (Chen et al., 2016) to simulate larger batch sizes and reduce memory footprint. Training is conducted in mixed bfloat16 precision (Micikevicius et al., 2017; Wang and Kanwar, 2019) with DeepSpeed's ZeRO-1 optimization (Rajbhandari et al., 2020), reducing memory consumption and allowing for larger model training. This comprehensive setup optimizes efficiency and performance, providing an effective solution for large-scale deep learning model training.

Our training was conducted on 4 NVIDIA DGX servers, each equipped with 8 A100 GPUs, and spanned a period of 4 weeks. Inference was performed on a single NVIDIA DGX server with eight A100 GPUs, utilizing parallel batch inference. During the pre-training phase, we employed a data packing strategy (Krell et al., 2021) and batch dynamic right padding strategy in the instruction tuning phase (Wolf et al., 2020), while the inference phase incorporated a batch left padding strategy (Wolf et al., 2020).

3.2 Evaluation Results

In this section, we provide a detailed analysis of the evaluation and results for both the English and Chinese benchmarks. We discuss task-specific performances and identify key areas of strengths and weaknesses for each model. The following sections present insights for the English and Chinese benchmarks, each highlighting the differences in model capabilities across a variety of NLP tasks.

In this overall analysis, we synthesize insights from both the English and Chinese benchmarks by evaluating the strengths and weaknesses of each model, as well as examining the performance across different types of tasks and datasets.

From the perspective of individual models in Figure.3, **GPT-4o** shows strong performance in sentiment analysis and structured tasks like multiple choice, indicating robust general language understanding capabilities. However, its weakness lies in relation extraction and detailed entity extraction, which require detailed understanding of complex financial relations. **FinMA-7B** stands out in sentiment tasks but lacks versatility, especially in question answering and summarization, likely due to the absence of targeted training for diverse NLP challenges. **Qwen-3-8B** demonstrates a balanced yet still improvable performance profile: it achieves decent results in sentiment analysis tasks but shows notable deficiencies in more complex tasks like question answering and summarization, indicating the need for more targeted domain adaptation training to enhance its specialized capabilities. **Llama-3-8B-Instruct** excels in english NLU tasks, but shows limitations in tasks requiring chinese tasks, such as entity and relation analysis. The metrics of **FinGPT-8B-lora** indicating that the current level of domain-specific tuning is insufficient for complex financial tasks. Finally, **DISC-FinLLM-Full** and **CFGPT1-7B-Full** demonstrate moderate strengths in entity extraction tasks but lack the robustness needed for broader NLP capabilities, revealing significant gaps in financial language comprehension.

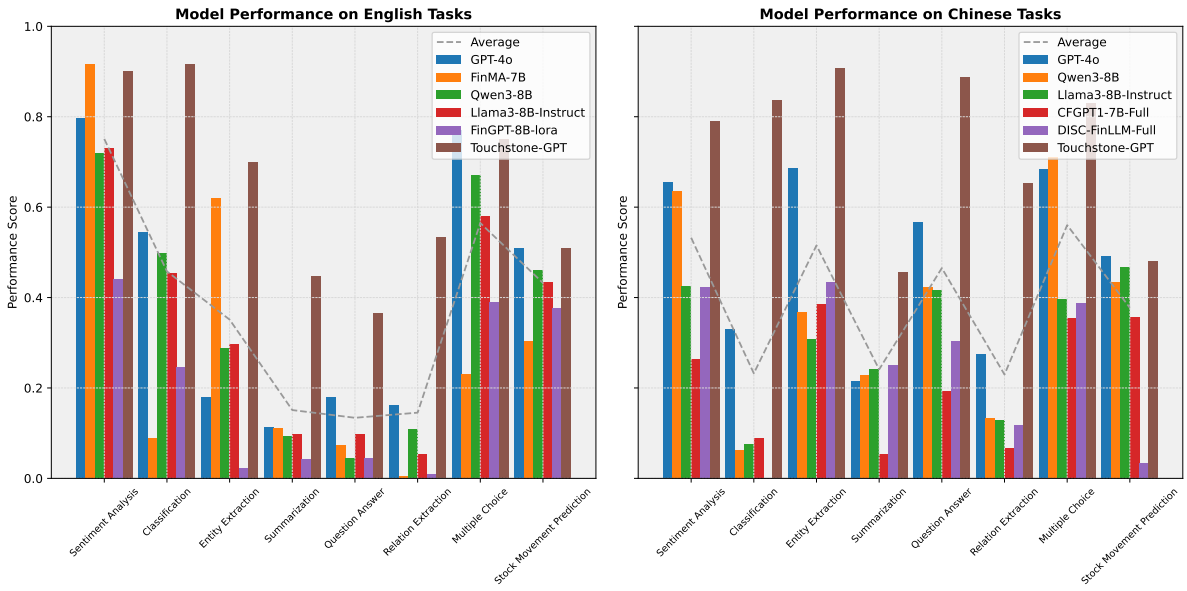

Figure 3: Comparison of performance from the perspective of models. Each subplot represents the performance of a models on both English and Chinese tasks. The bars indicate the model's performance on each task, while the dashed gray line represents the average performance across all models for that task.

From a task perspective in Figure.4, we observe that **Sentiment Analysis** generally yields high scores across most models, particularly for the English benchmark, indicating that sentiment understanding, even in financial contexts, is relatively well addressed by these models. In contrast, **Relation Extraction** and **Question Answer** in financial domain exhibit notably lower performance, especially for the Chinese benchmark. These results suggest that capturing financial relationships and classifying detailed financial statements pose greater challenges, requiring more sophisticated training datasets or better model architectures. The LendingClub dataset in **Classification** is a specialized dataset in the field of risk control, requiring more targeted fine-tuning to achieve good results. **Stock Movement Prediction** also shows low performance across most models, with only a few models such as **GPT-4o** demonstrating relatively moderate performance, but it is still practically unusable, highlighting the inherent difficulty of this task. Market prediction relying solely on news information is likely insufficient; volume-price data and factor analysis can provide more comprehensive information. However, current large language models are unable to process these inputs, which is a significant area of future research. **Summarization** also stands out as a weak area for most models, with consistently low BLEU and Rouge scores, reflecting the challenges in generating concise, coherent summaries of complex financial text.

Overall, the insights suggest that while models like **GPT-4o**, **FinMA-7B**, and **Touchstone-GPT** have particular strengths in sentiment analysis and some structured tasks, the overall capability to handle comprehensive financial NLP tasks remains limited. Most models require targeted improvements, especially for relation extraction, summarization, question answering and stock movement prediction in both English and Chinese contexts. This calls for more domain-specific training and the development of specialized datasets that focus on capturing the detailed and often complex financial language, which is crucial for advancing the performance of financial large language models. Furthermore, while **Touchstone-GPT** demonstrates competitive performance across various tasks due to its robust pre-training and instruction tuning, ongoing refinements and specialized tuning efforts are needed to address specific deficiencies observed in tasks such as summarization, relation extraction, question answering and stock movement prediction.

4 Related Works

4.1 Financial Large Language Models

In recent years, large language models (LLMs) tailored for the financial domain have gained significant attention. BloombertGPT (Wu et al., 2023) marked the beginning of the FinLLM era. Fin-

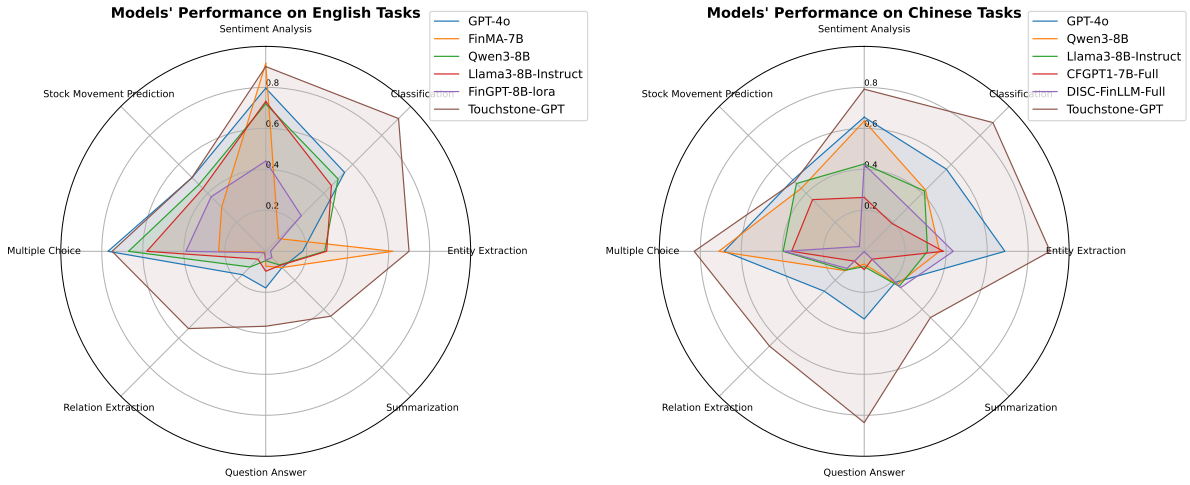

Figure 4: Comparison of performance from the perspective of tasks, illustrating average performance for English and Chinese tasks respectively.

Table 2: Performance metrics of financial large language models across English tasks like Sentiment Analysis, Classification, and Summarization. Models include GPT-4o, Llama-3-8B, Qwen-3-8B, FinMA-7B, FinGPT-8B, Mistral-7B, and Touchstone-GPT. The best results of each dataset are marked in **bold**.

Task	Dataset	Metrics	GPT-4o	FinMA-7B full	Qwen-3-8B	Llama-3-8B Instruct	FinGPT-8B lora	Mistral-7B	Touchstone GPT
Sentiment Analysis	FPB	Weighted-F1	0.8084	**0.9400**	0.8100	0.7631	0.2727	0.8110	0.8576
		ACC	0.8093	**0.9402**	0.8020	0.7660	0.3072	0.8030	0.8557
	Fiqa-SA	Weighted-F1	0.8106	0.8370	0.6800	0.7515	0.5885	0.6810	**0.8591**
		ACC	0.7702	0.8340	0.6000	0.7064	0.5872	0.6010	**0.8638**
Classification	Headlines	Weighted-F1	0.7857	0.9739	0.7300	0.7006	0.4516	0.7310	**0.9866**
		ACC	0.7931	0.9739	0.7270	0.7004	0.4331	0.7280	**0.9866**
	FOMC	Weighted-F1	0.6603	0.3988	0.6150	0.4904	0.2758	0.6160	**0.8788**
		ACC	0.6794	0.4274	0.6230	0.5625	0.2702	0.6240	**0.8790**
	lendingclub	Weighted-F1	0.6730	0.1477	0.5960	0.5943	0.5480	0.5970	**0.9783**
		MCC	0.1642	-0.6218	0.1720	0.1670	-0.1120	0.1730	**0.9297**
Entity Extraction	NER	Entity-F1	0.1800	0.6200	0.2900	0.2973	0.0231	0.2910	**0.6993**
Relation Extraction	FinRE	Relation-F1	0.1613	0.0054	0.1100	0.0540	0.0100	0.1110	**0.5331**
Multiple Choice	CFA	Weighted-F1	**0.7700**	0.2200	0.6720	0.5800	0.3993	0.6730	0.7497
		ACC	**0.7700**	0.2400	0.6720	0.5800	0.3800	0.6730	0.7500
Summarization	EDTSUM	Rouge-1	0.1675	0.1566	0.1480	0.1467	0.0622	0.1490	**0.5254**
		Rouge-2	0.0556	0.0491	0.0440	0.0429	0.0085	0.0450	**0.3446**
		Rouge-L	0.1069	0.1060	0.0860	0.0930	0.0412	0.0870	**0.4705**
		BLEU	0.1192	0.1361	0.1010	0.1085	0.0592	0.1020	**0.4512**
Question Answering	Finqa	RMACC	0.1037	0.0497	0.0280	0.0470	0.0110	0.0290	**0.2258**
	Convfinqa	RMACC	0.2540	0.0953	0.0650	0.1477	0.0772	0.0660	**0.5053**
Stock Movement Prediction	CMIN-US	Weighted-F1	0.5025	0.2639	0.4130	0.3722	0.3379	0.4140	**0.5036**
		ACC	**0.5149**	0.3446	0.5120	0.4955	0.4154	0.5130	0.5144

Table 3: Performance metrics of financial large language models across chinese tasks like Sentiment Analysis, Classification, and Summarization. Models include GPT-4o, Llama-3-8B, Qwen-3-8B, CFGPT-7B, DISC-FinLLM, and Touchstone-GPT. The best results of each dataset are marked in **bold**.

Task	Dataset	Metrics	GPT-4o	Qwen-3-8B Instruct	Llama-3-8B Instruct	CFGPT1-7B Full	DISC-FinLLM Full	Touchstone GPT
Sentiment Analysis	FinFe-CN	Weighted-F1	0.6593	0.6478	0.3633	0.2528	0.4177	**0.7888**
		ACC	0.6500	0.6632	0.4891	0.2732	0.4292	**0.7936**
Classification	FinNL-CN	ORMACC	0.3303	0.0824	0.0747	0.0894	0.0011	**0.8360**
Entity Extraction	FinESE-CN	ORMACC	0.6867	0.3879	0.3088	0.3863	0.4346	**0.9074**
Relation Extraction	FinRE-CN	RMACC	0.2754	0.1532	0.1296	0.0678	0.1182	**0.6541**
Multiple Choice	FinEval	Weighted-F1	**0.7364**	0.7332	0.4432	0.3543	0.4288	0.7361
		ACC	**0.7353**	0.7337	0.4471	0.3529	0.4294	**0.7353**
	CPA	Weighted-F1	0.6312	0.7159	0.3421	0.3543	0.3451	**0.9238**
		ACC	0.6309	0.7162	0.3504	0.3553	0.3518	**0.9238**
Summarization	FinNA-CN	Rouge-1	0.3197	0.3528	0.3477	0.1018	0.3486	**0.5526**
		Rouge-2	0.1434	0.1699	0.1702	0.0263	0.1678	**0.3603**
		Rouge-L	0.2511	0.2846	0.2802	0.0650	0.2997	**0.5214**
		BLEU	0.1423	0.1743	0.1672	0.0238	0.1885	**0.3944**
Question Answering	FinQa-CN	RMACC	0.6578	0.5245	0.4540	0.1126	0.3949	**0.9214**
	FinCQa-CN	RMACC	0.4765	0.3624	0.3787	0.2714	0.2134	**0.8552**
Stock Movement Prediction	CMIN-CN	Weighted-F1	**0.4858**	0.4165	0.4497	0.3549	0.0329	0.4735
		ACC	0.4988	0.4825	0.4858	0.3584	0.0332	**0.4878**

GPT (Yang et al., 2023) introduced an open-source framework emphasizing a data-centric approach with lightweight low-rank adaptation techniques. PIXIU (Xie et al., 2023) provided a comprehensive framework, presenting the first financial LLM fine-tuned on LLaMA with a 136K instruction dataset and evaluation benchmark. CFGPT (Li et al., 2023) developed a Chinese Financial Generative Pre-trained Transformer framework, encompassing dataset, model, and deployment capabilities. DISC-FinLLM (Chen et al., 2023) enhanced general LLMs through a multiple experts fine-tuning framework, expanding domain-specific capabilities.

4.2 Benchmarks for FinLLMs

In the current landscape of open-source financial evaluation benchmarks, FLUE (Shah et al., 2022) was the first open-source benchmark introduced for financial language understanding in English. It includes five financial tasks: Financial Sentiment Analysis (Malo et al., 2014; Maia et al., 2018a), Financial News Classification (Sinha and Khandait, 2021), Structure Boundary Detection (FinSBD3, 2021), Named Entity Recognition (Alvarado et al., 2015), and Question Answering (Maia et al., 2018b). Building on FLUE, FinGPT (Yang et al., 2023) introduced the Financial Relation Extraction task with the addition of the FinRED dataset (Sharma et al., 2022), while also expanding the sentiment analysis task by incorporating TFNS (Magic, 2022) and NWGI (Yang, 2023) datasets. Additionally, FinGPT created prompt-based instructions across all datasets, culminating in the Financial Instruction Tuning Benchmark. Around the same time, other open-source benchmarks, such as PIXIU (Xie et al., 2023) and FinBen (Xie et al., 2024), were developed. PIXIU includes a wide range of financial tasks and datasets, while FinBen, expanding on PIXIU, introduced numerous supplementary financial datasets to evaluate LLMs comprehensively and gauge their proficiency across various financial scenarios.In the Chinese domain, BBT-Benchmark (Lu et al., 2023) was the first Chinese financial benchmark, covering multiple financial NLP tasks, including sentiment analysis, news classification, summarization, relation extraction, and question answering. Later, FinEval (Zhang et al., 2023) and FinanceIQ (Zhang and Yang, 2023)benchmark emerged, providing an evaluation framework for LLMs in financial contexts, encompassing multiple-choice questions across finance, economics, accounting, and certification. DISC-FinLLM (Chen et al., 2023) evaluated financial LLMs on top of these benchmarks, introducing additional tasks focused on financial calculations and current events to assess LLMs in applied financial scenarios. The most recent Chinese Financial benchmark, CFBenchmark (Lei et al., 2023), introduced three key tasks like Financial Entity Recognition, expanding evaluation resources for Chinese financial tasks.

The increasing number of benchmarks in both English and Chinese financial domains has established valuable evaluation frameworks for LLMs

in finance, though challenges remain. For example, data quality across various English financial datasets is inconsistent, limiting LLMs from reaching their full potential, as seen in FinBen and PIXIU, where low-quality datasets yielded suboptimal performance metrics. In Chinese benchmarks, tasks are often skewed toward multiple-choice questions, with limited alignment with English financial tasks. Moreover, no bilingual benchmark currently exists for comprehensive LLM evaluation in financial NLP tasks across English and Chinese. In this work, we integrate representative datasets from both languages, creating FinLLM-benchmark. We compile and align eight financial NLP tasks in English and Chinese, selecting high-quality datasets for each task and appropriate metrics to effectively assess model performance.

5 Conclusion

In this study, we introduce the Golden Touchstone benchmark, the inaugural structured and comprehensive bilingual benchmark specifically designed for English-Chinese financial NLP. This benchmark encompasses a wide array of financial NLP tasks, including Natural Language Understanding (NLU) and Natural Language Generation (NLG) across eight categories: Sentiment Analysis, Classification, Entity Extraction, Summarization, Stock Market Prediction, Question Answering, Relation Extraction, and Multiple Choice. By leveraging existing high-quality open-source financial datasets, we curated representative datasets and selected appropriate evaluation metrics for each task category. Utilizing these resources, we conducted extensive evaluations of current models such as GPT-4o and prominent open-source financial LLMs, including FinGPT and FinMA, thereby establishing performance benchmarks for financial LLMs within bilingual contexts. Moreover, we contributed to the community by open-sourcing Touchstone-GPT, a robust financial LLM that employs a two-stage training approach and has demonstrated superior input-based inference capabilities on the Golden Touchstone benchmark compared to GPT-4o. Our open-source initiative provides a bilingual English-Chinese evaluation framework aimed at fostering the sustainable development of LLMs in a multilingual financial environment.

6 Limitations

Despite these advancements, the benchmark currently exhibits certain limitations, including a limited range of NLG tasks and a focus solely on single-modality. Future enhancements will include the integration of additional NLG tasks, such as extended text generation for financial report analysis and more sophisticated sentiment assessments. Dataset biases and model scalability are also challenges. Furthermore, we plan to expand the benchmark to cover other financial sectors such as insurance and futures trading, thus broadening the scope and applicability of financial LLM assessments across diverse scenarios. Also, the performance of Touchstone-GPT on specific tasks within the Golden Touchstone benchmark, particularly in stock market prediction, requires further improvement. Our subsequent research will explore the incorporation of agent-based and retrieval-augmented generation (RAG) methods to augment the model's capabilities in numerical computation and real-time news analysis. Additionally, we aim to venture into multimodal modeling, integrating visual data and time-series data for tasks such as financial time-series forecasting, financial chart analysis, and content generation.

7 Ethical Statements

We do not see our work to have possible harmful outcomes. We follow the ACL ethical guidelines when conducting the research in this paper.

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A Overview of Finance Evaluation Datasets by Task Type, Sample Sizes (Training, Validation, Test), and Evaluation Metrics

In the Table.4 and 5, we show the overview of finance evaluation datasets by task type, sample sizes (training, validation, test), evaluation metrics and license.

B Additional Experiment Results

Due to computational constraints imposed by our large-scale dataset (more than 112K samples; see Appendix A) and to ensure a fair comparison among models of similar scale, the main paper (Table 2 and Table 3) primarily reports results from models with <10B parameters, alongside GPT-4o. In this appendix, we supplement those findings with evaluations of several advanced or larger models (>100B parameters) in Table.6, providing a more comprehensive performance comparison. These models include Yi-34B(Young et al., 2024), GLM-4-32B-0414(GLM et al., 2024), Mixtral 8*7B(Jiang et al., 2024), Qwen2.5-32B/72B(Yang et al., 2024a), Xuanyuan 2.0(Zhang and Yang, 2023) and deepseek-v3(Liu et al., 2024). Furthermore, we extended the CMIN dataset to introduce Stock-CoT-R1 [1], a synthetic dataset for stock prediction tasks.

C Inference Template of Large Language Models

The Table.7 showcases how inference templates vary across different models. It is crucial to select the appropriate template for constructing correct inputs when inferring on the test sets of datasets. An incorrect template can significantly impair the performance of a model. We have observed that the underperformance of some large financial language models in some benchmarks is precisely due to not selecting the appropriate templates for evaluation. For more details of training and inference template, please refer to our open-source code repository on Github.

D Typical Case Study Analysis of Typical Financial NLP Tasks

This appendix provides a detailed case study analysis for some typical financial NLP tasks: financial sentiment analysis, text classification, entity extraction, and stock movement prediction. Each analysis is presented in a separate table, categorizing data sets, instructions, inputs, labels, and predictions from multiple models.

Financial Sentiment Analysis As demonstrated in Table 8, financial sentiment classification is one of the simpler tasks for benchmarks in financial NLP, resulting in high performance across all models tested. General-purpose models (GPT-4o, Qwen-3, Llama-3) provide not only the answer but also a detailed analysis, despite not being specifically fine-tuned on the FiQA-SA dataset. In contrast, specialized models (FinGPT, FinMA, Touchstone-GPT) that have undergone instruction tuning deliver straightforward, direct responses, illustrating their efficiency and focus in domain-specific applications.

Credit Rating Analysis In the classification task using the LendingClub dataset, which poses a challenging credit rating task, the models face a complex array of professional financial information evident in the input fields. Consequently, most models do not perform optimally. Among general models, GPT-4o exhibits the best performance, demon-

[1] https://huggingface.co/datasets/IDEA-FinAI/Finance-R1-Reasoning

Table 4: Overview of English Finance Evaluation Datasets by Task Type, Sample Sizes (Training, Validation, Test), Evaluation Metrics and License

Task	Dataset	Train	Valid	Test	Metrics	License
Sentiment Analysis	**FPB**	3100	776	970	Weighted-F1 ACC	CC BY-SA 3.0
	FiQA-SA	750	188	235	Weighted-F1 ACC	MIT License
Classification	**Headlines**	71900	10300	20500	Weighted-F1 ACC	CC BY-SA 3.0
	FOMC	1984	-	496	Weighted-F1 ACC	CC BY-SA 3.0
	lendingclub	9417	1345	2691	Weighted-F1 MCC	
Entity Recognition	**NER**	408	103	98	Entity-F1	CC BY-SA 3.0
Relation Extraction	**FinRE**	27558	-	5112	Relation-F1	MIT License
Multiple Choice	**CFA**	1884	100	20	Weighted-F1 ACC	CC BY-SA 3.0
Summarization	**EDTSUM**	8000	-	2000	ROUGE BLEU	MIT License
Question Answering	**FinQa**	6251	883	1147	RMACC	MIT License
	ConvfinQa	8890	2210	1490	RMACC	MIT License
Stock Movement Prediction	**CMIN-US**	88297	9010	8480	Weighted-F1 ACC	MIT License

Table 5: Overview of Chinese Finance Evaluation Datasets by Task Type, Sample Sizes (Training, Validation, Test), and Evaluation Metrics

Task	Dataset	Train	Valid	Test	Metrics	License
Sentiment Analysis	**FinFE-CN**	16157	2020	2020	Weighted-F1 ACC	Public
Classification	**FinNL-CN**	7071	884	884	ORMACC	Public
Entity Extraction	**FinESE-CN**	14252	1781	1782	ORMACC	Public
Relation Extraction	**FinRE-CN**	13486	1489	3727	RMACC	Public
Multiple Choice	**FinEval**	1071	170	3340	Weighted-F1 ACC	Apache-2.0 license
	CPA	6268	1444	6	Weighted-F1 ACC	CC BY-SA 3.0
Summarization	**FinNA-CN**	28800	3600	3600	ROUGE BLEU	Public
Question Answering	**FinQa-CN**	19906	2469	2480	RMACC	Public
	FincQa-CN	21965	2741	2745	RMACC	Public
Stock Movement Prediction	**CMIN-CN**	214873	23904	23571	Weighted-F1 ACC	MIT License

Table 6: Extended performance metrics of financial large language models with additional models and datasets. Best results are marked in **bold**.

Task	Dataset	Metrics	GPT-4o	Yi-34B	GLM-4-32B -0414	Mixtral 8x7B	Qwen2.5 32B	Qwen2.5 72B	Xuanyuan 2.0	deepseek -v3	Touchstone GPT
Sentiment Analysis	FPB	Weighted-F1	0.8084	0.8652	0.8921	0.8510	0.8783	0.8815	0.8426	**0.9014**	0.8576
		ACC	0.8093	0.8638	0.8905	0.8492	0.8761	0.8798	0.8403	**0.8997**	0.8557
	Fiqa-SA	Weighted-F1	0.8106	0.8321	0.8453	0.8257	0.8389	0.8421	0.8185	0.8512	**0.8591**
		ACC	0.7702	0.8289	0.8402	0.8204	0.8337	0.8376	0.8123	0.8465	**0.8638**
Classification	Headlines	Weighted-F1	0.7857	0.9521	0.9623	0.9387	0.9589	0.9610	0.9324	0.9684	**0.9866**
		ACC	0.7931	0.9518	0.9619	0.9375	0.9582	0.9607	0.9312	0.9679	**0.9866**
	FOMC	Weighted-F1	0.6603	0.7824	0.8125	0.7653	0.8021	0.8089	0.7527	0.8237	**0.8788**
		ACC	0.6794	0.7852	0.8183	0.7716	0.8079	0.8135	0.7591	0.8294	**0.8790**
	lendingclub	Weighted-F1	0.6730	0.8927	0.9215	0.8753	0.9128	0.9186	0.8624	0.9357	**0.9783**
		MCC	0.1642	0.7825	0.8432	0.7521	0.8279	0.8375	0.7326	0.8674	**0.9297**
Entity Extraction	NER	Entity-F1	0.1800	0.4521	0.5832	0.4125	0.5527	0.5718	0.3924	0.6127	**0.6993**
Relation Extraction	FinRE	Relation-F1	0.1613	0.2854	0.3726	0.2543	0.3428	0.3615	0.2327	0.4128	**0.5331**
Multiple Choice	CFA	Weighted-F1	**0.7700**	0.6521	0.7125	0.6234	0.6928	0.7036	0.6057	0.7289	0.7497
		ACC	**0.7700**	0.6528	0.7132	0.6241	0.6935	0.7043	0.6064	0.7296	0.7500
Summarization	EDTSUM	Rouge-1	0.1675	0.3521	0.4128	0.3276	0.3987	0.4089	0.3124	0.4287	**0.5254**
		Rouge-2	0.0556	0.2187	0.2875	0.2013	0.2739	0.2832	0.1924	0.3057	**0.3446**
		Rouge-L	0.1069	0.2854	0.3521	0.2678	0.3389	0.3476	0.2543	0.3724	**0.4705**
		BLEU	0.1192	0.3025	0.3687	0.2854	0.3562	0.3641	0.2728	0.3865	**0.4512**
Question Answering	Finqa	RMACC	0.1037	0.1682	0.1925	0.1543	0.1837	0.1891	0.1476	0.2014	**0.2258**
	Convfinqa	RMACC	0.2540	0.3827	0.4216	0.3652	0.4089	0.4173	0.3524	0.4385	**0.5053**
Stock Movement Prediction	CMIN-CoT-R1	Weighted-F1	0.5025	0.4873	0.4982	0.4791	0.4924	0.4957	0.4738	0.5011	**0.5136**
		ACC	0.5149	0.5027	0.5103	0.4975	0.5068	0.5092	0.4921	0.5138	**0.5224**

Table 7: Comparison of Inference Templates Across Different Models for Dataset Evaluation

Model	Template
GPT-4o	"<|im_start|>system{{system_prompt}}<|im_end|>\n" "<|im_start|>user{{instruction}}{{input}}<|im_end|>\n" "<|im_start|>assistant\n"
Qwen-3	"<|im_start|>system{{system_prompt}}<|im_end|>\n" "<|im_start|>user{{instruction}}{{input}}<|im_end|>\n" "<|im_start|>assistant\n"
Llama-3	"<|start_header_id|>system<|end_header_id|>" "{{system_prompt}}<|eot_id|>\n" "<|start_header_id|>user<|end_header_id|>" "{{instruction}}{{input}}<|eot_id|>\n" "<|start_header_id|>assistant<|end_header_id|>\n"
FinGPT	"Instruction:{{instruction}}" "Input{{input}}\nAnswer:"
FinMA	"Human:{{instruction}}{{input}}\n" "Assistant:\n"
CFGPT	"{{instruction}}{{input}}\n"
DISC-FinLLM	"<reserved_102> {{instruction}}{{input}}<reserved_103>"
Touchstone	"<|im_start|>system{{system_prompt}}<|im_end|>\n" "<|im_start|>user{{instruction}}{{input}}<|im_end|>\n" "<|im_start|>assistant\n"

Table 8: Detailed Case Study Analysis of Financial Sentiment Analysis on the FiQA-SA dataset, Financial Text Classification on the LendingClub dataset, Financial Entity Extraction on NER dataset, Stock Movement Prediction on DJIA dataset.

Dataset	Instruction	Input	Label	Model	Prediction
FiQA-SA	What is the sentiment of the following financial post: Positive, Negative, or Neutral?	@Joerazorback This time I think $AAPL will miss again. Generally speaking market is near top when everyone thinks that market goes only up.	Negative	GPT-4o	The sentiment of this financial post is **Negative**. The user expresses doubt about Apple's performance, suggesting that they think the stock will miss expectations. Additionally, the phrase about the market being "near top" implies a bearish outlook, indicating that the market may decline after its current uptrend.
				Qwen-3	The sentiment of the given financial post is Negative. The user is predicting that Apple's stock price ($AAPL) will miss or fail to meet expectations, and they also suggest that the market might be nearing its peak when everyone is optimistic about continuous upward movement.
				Llama-3	The sentiment of this post is Negative. The author is predicting that Apple's stock (AAPL) will miss expectations, which is a negative view. Additionally, the author is suggesting that the market is near a top, implying that a downturn is likely, which is also a negative sentiment.
				FinGPT	Negative
				FinMA	Negative
				Touchstone GPT	Negative
LendingClub	Assess the client's loan status based on the following loan records from Lending Club. Respond with only 'good' or 'bad'.	The client has attributes as follows: ... The state of Employment Length is 2 years. The state of Home Ownership is RENT. The state of Verification Status is Verified. The state of Annual Income is 83000.0.	Good	GPT-4o	Good
				Qwen-3	Good
				Llama-3	Good
				FinGPT	Bad
				FinMA	Bad
				Touchstone GPT	Good
NER	In the sentences extracted from financial agreements in U.S. SEC filings, identify the named entities.	WHEREAS: (A) The Borrower intends to develop a plant in France for the production of solar grade silicon.	Borrower, PER; France, LOC	GPT-4o	France, LOC
				Qwen-3	Borrower, PER; France, LOC; solar grade silicon, ORG
				Llama-3	Borrower, PER; France, LOC
				FinGPT	The Borrower intends to develop a plant in France for the production of solar grade silicon
				FinMA	Borrower, PER; France, LOC
				Touchstone GPT	Borrower, PER; France, LOC
CMIN-US	Based on the news about the stock in the past five trading days.....	NEWS1: Biden Urges CEOs to Improve U.S. Cybersecurity After Attacks...	0	GPT-4o	Rise
				Qwen3	Decrease
				Llama-3	Decrease
				FinGPT	""
				FinMA	Rise
				Touchstone GPT	Rise

strating the capabilities of large-scale models. In the realm of specialized financial language models, Touchstone-GPT, with its high-quality instruction tuning, significantly outperforms FinMA and FinGPT, which are only minimally tuned with Lora.

Financial NER In this information extraction task, most models demonstrated an understanding of the task intent and adhered to the instructions, signifying that even in the era of large language models, models like Qwen-3 and Llama-3 actually outperformed GPT-4o. In particular, specialized models such as FinMA and Touchstone-GPT, with more comprehensive instruction tuning, responded accurately and succinctly, highlighting their enhanced capability and focus on domain-specific tasks.

Stock Movement Prediction The Stock Movement Prediction task is one of the most challenging tasks, as it requires models to predict the daily fluctuations of the CMIN-US based solely on the 5-day news items. From the results in Table.4, it is evident that GPT-4o performed the best, yet it still falls short of practical utility. Even Touchstone-GPT, despite specialized instruction tuning, performed poorly. Our analysis suggests that the sentiment of news items may not reliably predict stock movements and that incorporating quantitative data is essential for achieving practical model performance. Similar conclusions were drawn from experiments with traditional machine learning methods like XGBoost. Nevertheless, aside from simpler tasks like financial sentiment analysis, we also pose challenging tasks such as stock prediction, which are closer to real-world applications, leaving more room for benchmark challenges and exploration. Multimodal fusion of news and quantitative data represents a promising future direction, and we look forward to seeing models excel in these tasks.

Due to the similar performance of models across corresponding task types on the Chinese benchmark, we will not reiterate these comparisons and analysis here.

E Information About Use Of Ai Assistants

In the preparation of this work, the author used AI-assisted technology (specifically, large language models such as GPT-4 and Deepseek-V3) exclusively for text refinement purposes. The AI was employed to assist in proofreading, correcting grammatical errors, and polishing linguistic expressions to improve the clarity and readability of the manuscript.